UM10360

Chapter 6: LPC176x/5x Nested Vectored Interrupt Controller (NVIC)

Rev. 4. 1 — 19 December 2016 **User manual**

6.1 Features

- Nested Vectored Interrupt Controller that is an integral part of the ARM Cortex-M3
- Tightly coupled interrupt controller provides low interrupt latency
- Controls system exceptions and peripheral interrupts
- In the LPC176x/5x, the NVIC supports 35 vectored interrupts
- 32 programmable interrupt priority levels, with hardware priority level masking
- Relocatable vector table
- Non-Maskable Interrupt
- Software interrupt generation

6.2 Description

The Nested Vectored Interrupt Controller (NVIC) is an integral part of the Cortex-M3. The tight coupling to the CPU allows for low interrupt latency and efficient processing of late arriving interrupts.

Refer to the Cortex-M3 User Guide Section 34.4.2 for details of NVIC operation.

6.3 Interrupt sources

Table 50 lists the interrupt sources for each peripheral function. Each peripheral device may have one or more interrupt lines to the Vectored Interrupt Controller. Each line may represent more than one interrupt source, as noted.

Exception numbers relate to where entries are stored in the exception vector table. Interrupt numbers are used in some other contexts, such as software interrupts.

In addition, the NVIC handles the Non-Maskable Interrupt (NMI). In order for NMI to operate from an external signal, the NMI function must be connected to the related device pin (P2.10 / EINT0n / NMI). When connected, a logic 1 on the pin will cause the NMI to be processed. For details, refer to the Cortex-M3 User Guide that is an appendix to this User Manual.

Table 50. Connection of interrupt sources to the Vectored Interrupt Controller

Interrupt ID	Exception Number	Vector Offset	Function	Flag(s)
0	16	0x40	WDT	Watchdog Interrupt (WDINT)
1	17	0x44	Timer 0	Match 0 - 1 (MR0, MR1) Capture 0 - 1 (CR0, CR1)
2	18	0x48	Timer 1	Match 0 - 2 (MR0, MR1, MR2) Capture 0 - 1 (CR0, CR1)
3	19	0x4C	Timer 2	Match 0-3 Capture 0-1
4	20	0x50	Timer 3	Match 0-3 Capture 0-1
5	21	0x54	UART0	Rx Line Status (RLS) Transmit Holding Register Empty (THRE) Rx Data Available (RDA) Character Time-out Indicator (CTI) End of Auto-Baud (ABEO) Auto-Baud Time-Out (ABTO)
6	22	0x58	UART1	Rx Line Status (RLS) Transmit Holding Register Empty (THRE) Rx Data Available (RDA) Character Time-out Indicator (CTI) Modem Control Change End of Auto-Baud (ABEO) Auto-Baud Time-Out (ABTO)
7	23	0x5C	UART 2	Rx Line Status (RLS) Transmit Holding Register Empty (THRE) Rx Data Available (RDA) Character Time-out Indicator (CTI) End of Auto-Baud (ABEO) Auto-Baud Time-Out (ABTO)
8	24	0x60	UART 3	Rx Line Status (RLS) Transmit Holding Register Empty (THRE) Rx Data Available (RDA) Character Time-out Indicator (CTI) End of Auto-Baud (ABEO) Auto-Baud Time-Out (ABTO)
9	25	0x64	PWM1	Match 0 - 6 of PWM1 Capture 0-1 of PWM1
10	26	0x68	I^2C0	SI (state change)
11	27	0x6C	I^2C1	SI (state change)
12	28	0x70	I^2C2	SI (state change)
13	29	0x74	SPI	SPI Interrupt Flag (SPIF) Mode Fault (MODF)

Table 50. Connection of interrupt sources to the Vectored Interrupt Controller

Interrupt ID	Exception Number	Vector Offset	Function	Flag(s)
14	30	0x78	SSP0	Tx FIFO half empty of SSP0 Rx FIFO half full of SSP0 Rx Timeout of SSP0 Rx Overrun of SSP0
15	31	0x7C	SSP 1	Tx FIFO half empty Rx FIFO half full Rx Timeout Rx Overrun
16	32	0x80	PLL0 (Main PLL)	PLL0 Lock (PLOCK0)
17	33	0x84	RTC	Counter Increment (RTCCIF) Alarm (RTCALF)
18	34	0x88	External Interrupt	External Interrupt 0 (EINT0)
19	35	0x8C	External Interrupt	External Interrupt 1 (EINT1)
20	36	0x90	External Interrupt	External Interrupt 2 (EINT2)
21	37	0x94	External Interrupt	External Interrupt 3 (EINT3). **Note:** EINT3 channel is shared with GPIO interrupts
22	38	0x98	ADC	A/D Converter end of conversion
23	39	0x9C	BOD	Brown Out detect
24	40	0xA0	USB	USB_INT_REQ_LP, USB_INT_REQ_HP, USB_INT_REQ_DMA
25	41	0xA4	CAN	CAN Common, CAN 0 Tx, CAN 0 Rx, CAN 1 Tx, CAN 1 Rx
26	42	0xA8	GPDMA	IntStatus of DMA channel 0, IntStatus of DMA channel 1
27	43	0xAC	I^{2}S	irq, dmareq1, dmareq2
28	44	0xB0	Ethernet	WakeupInt, SoftInt, TxDoneInt, TxFinishedInt, TxErrorInt, TxUnderrunInt, RxDoneInt, RxFinishedInt, RxErrorInt, RxOverrunInt.
29	45	0xB4	Repetitive Interrupt Timer	RITINT
30	46	0xB8	Motor Control PWM	IPER[2:0], IPW[2:0], ICAP[2:0], FES
31	47	0xBC	Quadrature Encoder	INX_Int, TIM_Int, VELC_Int, DIR_Int, ERR_Int, ENCLK_Int, POS0_Int, POS1_Int, POS2_Int, REV_Int, POS0REV_Int, POS1REV_Int, POS2REV_Int
32	48	0xC0	PLL1 (USB PLL)	PLL1 Lock (PLOCK1)
33	49	0xC4	USB Activity Interrupt	USB_NEED_CLK
34	50	0xC8	CAN Activity Interrupt	CAN1WAKE, CAN2WAKE

6.4 Vector table remapping

The Cortex-M3 incorporates a mechanism that allows remapping the interrupt vector table to alternate locations in the memory map. This is controlled via the Vector Table Offset Register (VTOR) contained in the Cortex-M3.

The vector table may be located anywhere within the bottom 1 GB of Cortex-M3 address space. The vector table should be located on a 256 word (1024 byte) boundary to insure alignment on LPC176x/5x family devices. Refer to Section 34.4.3.5 of the Cortex-M3 User Guide appended to this manual for details of the Vector Table Offset feature.

ARM describes bit 29 of the VTOR (TBLOFF) as selecting a memory region, either code or SRAM. For simplicity, this bit can be thought as simply part of the address offset since the split between the “code” space and the “SRAM” space occurs at the location corresponding to bit 29 in a memory address.

Examples:

To place the vector table at the beginning of the “local” static RAM, starting at address 0x1000 0000, place the value 0x1000 0000 in the VTOR register. This indicates address 0x1000 0000 in the code space, since bit 29 of the VTOR equals 0.

To place the vector table at the beginning of the AHB static RAM, starting at address 0x2007 C000, place the value 0x2007 C000 in the VTOR register. This indicates address 0x2007 C000 in the SRAM space, since bit 29 of the VTOR equals 1.

6.5 Register description

The following table summarizes the registers in the NVIC as implemented in the LPC176x/5x. The Cortex-M3 User Guide Section 34.4.2 provides a functional description of the NVIC.

Table 51. NVIC register map

Name	Description	Access	Reset value	Address
ISER0 to ISER1	Interrupt Set-Enable Registers. These 2 registers allow enabling interrupts and reading back the interrupt enables for specific peripheral functions.	RW	0	ISER0 - 0xE000 E100 ISER1 - 0xE000 E104
ICER0 to ICER1	Interrupt Clear-Enable Registers. These 2 registers allow disabling interrupts and reading back the interrupt enables for specific peripheral functions.	RW	0	ICER0 - 0xE000 E180 ICER1 - 0xE000 E184
ISPR0 to ISPR1	Interrupt Set-Pending Registers. These 2 registers allow changing the interrupt state to pending and reading back the interrupt pending state for specific peripheral functions.	RW	0	ISPR0 - 0xE000 E200 ISPR1 - 0xE000 E204
ICPR0 to ICPR1	Interrupt Clear-Pending Registers. These 2 registers allow changing the interrupt state to not pending and reading back the interrupt pending state for specific peripheral functions.	RW	0	ICPR0 - 0xE000 E280 ICPR1 - 0xE000 E284
IABR0 to IABR1	Interrupt Active Bit Registers. These 2 registers allow reading the current interrupt active state for specific peripheral functions.	RO	0	IABR0 - 0xE000 E300 IABR1 - 0xE000 E304
IPR0 to IPR8	Interrupt Priority Registers. These 9 registers allow assigning a priority to each interrupt. Each register contains the 5-bit priority fields for 4 interrupts.	RW	0	IPR0 - 0xE000 E400 IPR1 - 0xE000 E404 IPR2 - 0xE000 E408 IPR3 - 0xE000 E40C IPR4 - 0xE000 E410 IPR5 - 0xE000 E414 IPR6 - 0xE000 E418 IPR7 - 0xE000 E41C IPR8 - 0xE000 E420
STIR	Software Trigger Interrupt Register. This register allows software to generate an interrupt.	WO	0	STIR - 0xE000 EF00

6.5.1 Interrupt Set-Enable Register 0 register (ISER0 - 0xE000 E100)

The ISER0 register allows enabling the first 32 peripheral interrupts, or for reading the enabled state of those interrupts. The remaining interrupts are enabled via the ISER1 register (Section 6.5.2). Disabling interrupts is done through the ICER0 and ICER1 registers (Section 6.5.3 and Section 6.5.4).

Table 52. Interrupt Set-Enable Register 0 register (ISER0 - 0xE000 E100)

Bit	Name	Function
0	ISE_WDT	Watchdog Timer Interrupt Enable. Write: writing 0 has no effect, writing 1 enables the interrupt. Read: 0 indicates that the interrupt is disabled, 1 indicates that the interrupt is enabled.
1	ISE_TIMER0	Timer 0 Interrupt Enable. See functional description for bit 0.
2	ISE_TIMER1	Timer 1. Interrupt Enable. See functional description for bit 0.
3	ISE_TIMER2	Timer 2 Interrupt Enable. See functional description for bit 0.
4	ISE_TIMER3	Timer 3 Interrupt Enable. See functional description for bit 0.
5	ISE_UART0	UART0 Interrupt Enable. See functional description for bit 0.
6	ISE_UART1	UART1 Interrupt Enable. See functional description for bit 0.
7	ISE_UART2	UART2 Interrupt Enable. See functional description for bit 0.
8	ISE_UART3	UART3 Interrupt Enable. See functional description for bit 0.
9	ISE_PWM	PWM1 Interrupt Enable. See functional description for bit 0.
10	ISE_I2C0	I^2C0 Interrupt Enable. See functional description for bit 0.
11	ISE_I2C1	I^2C1 Interrupt Enable. See functional description for bit 0.
12	ISE_I2C2	I^2C2 Interrupt Enable. See functional description for bit 0.
13	ISE_SPI	SPI Interrupt Enable. See functional description for bit 0.
14	ISE_SSP0	SSP0 Interrupt Enable. See functional description for bit 0.
15	ISE_SSP1	SSP1 Interrupt Enable. See functional description for bit 0.
16	ISE_PLL0	PLL0 (Main PLL) Interrupt Enable. See functional description for bit 0.
17	ISE_RTC	Real Time Clock (RTC) Interrupt Enable. See functional description for bit 0.
18	ISE_EINT0	External Interrupt 0 Interrupt Enable. See functional description for bit 0.
19	ISE_EINT1	External Interrupt 1 Interrupt Enable. See functional description for bit 0.
20	ISE_EINT2	External Interrupt 2 Interrupt Enable. See functional description for bit 0.
21	ISE_EINT3	External Interrupt 3 Interrupt Enable. See functional description for bit 0.
22	ISE_ADC	ADC Interrupt Enable. See functional description for bit 0.
23	ISE_BOD	BOD Interrupt Enable. See functional description for bit 0.
24	ISE_USB	USB Interrupt Enable. See functional description for bit 0.
25	ISE_CAN	CAN Interrupt Enable. See functional description for bit 0.
26	ISE_DMA	GPDMA Interrupt Enable. See functional description for bit 0.
27	ISE_I2S	I^2S Interrupt Enable. See functional description for bit 0.
28	ISE_ENET	Ethernet Interrupt Enable. See functional description for bit 0.
29	ISE_RIT	Repetitive Interrupt Timer Interrupt Enable. See functional description for bit 0.
30	ISE_MCPWM	Motor Control PWM Interrupt Enable. See functional description for bit 0.
31	ISE_QEI	Quadrature Encoder Interface Interrupt Enable. See functional description for bit 0.

6.5.2 Interrupt Set-Enable Register 1 register (ISER1 - 0xE000 E104)

The ISER1 register allows enabling the second group of peripheral interrupts, or for reading the enabled state of those interrupts. Disabling interrupts is done through the ICER0 and ICER1 registers (Section 6.5.3 and Section 6.5.4).

Table 53. Interrupt Set-Enable Register 1 register (ISER1 - 0xE000 E104)

Bit	Name	Function
0	ISE_PLL1	PLL1 (USB PLL) Interrupt Enable. Write: writing 0 has no effect, writing 1 enables the interrupt. Read: 0 indicates that the interrupt is disabled, 1 indicates that the interrupt is enabled.
1	ISE_USBACT	USB Activity Interrupt Enable. See functional description for bit 0.
2	ISE_CANACT	CAN Activity Interrupt Enable. See functional description for bit 0.
31:3	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.

6.5.3 Interrupt Clear-Enable Register 0 (ICER0 - 0xE000 E180)

The ICER0 register allows disabling the first 32 peripheral interrupts, or for reading the enabled state of those interrupts. The remaining interrupts are disabled via the ICER1 register (Section 6.5.4). Enabling interrupts is done through the ISER0 and ISER1 registers (Section 6.5.1 and Section 6.5.2).

Table 54. Interrupt Clear-Enable Register 0 (ICER0 - 0xE000 E180)

Bit	Name	Function
0	ICE_WDT	Watchdog Timer Interrupt Disable. Write: writing 0 has no effect, writing 1 disables the interrupt. Read: 0 indicates that the interrupt is disabled, 1 indicates that the interrupt is enabled.
1	ICE_TIMER0	Timer 0 Interrupt Disable. See functional description for bit 0.
2	ICE_TIMER1	Timer 1. Interrupt Disable. See functional description for bit 0.
3	ICE_TIMER2	Timer 2 Interrupt Disable. See functional description for bit 0.
4	ICE_TIMER3	Timer 3 Interrupt Disable. See functional description for bit 0.
5	ICE_UART0	UART0 Interrupt Disable. See functional description for bit 0.
6	ICE_UART1	UART1 Interrupt Disable. See functional description for bit 0.
7	ICE_UART2	UART2 Interrupt Disable. See functional description for bit 0.
8	ICE_UART3	UART3 Interrupt Disable. See functional description for bit 0.
9	ICE_PWM	PWM1 Interrupt Disable. See functional description for bit 0.
10	ICE_I2C0	I^2C0 Interrupt Disable. See functional description for bit 0.
11	ICE_I2C1	I^2C1 Interrupt Disable. See functional description for bit 0.
12	ICE_I2C2	I^2C2 Interrupt Disable. See functional description for bit 0.
13	ICE_SPI	SPI Interrupt Disable. See functional description for bit 0.
14	ICE_SSP0	SSP0 Interrupt Disable. See functional description for bit 0.
15	ICE_SSP1	SSP1 Interrupt Disable. See functional description for bit 0.
16	ICE_PLL0	PLL0 (Main PLL) Interrupt Disable. See functional description for bit 0.
17	ICE_RTC	Real Time Clock (RTC) Interrupt Disable. See functional description for bit 0.
18	ICE_EINT0	External Interrupt 0 Interrupt Disable. See functional description for bit 0.
19	ICE_EINT1	External Interrupt 1 Interrupt Disable. See functional description for bit 0.
20	ICE_EINT2	External Interrupt 2 Interrupt Disable. See functional description for bit 0.
21	ICE_EINT3	External Interrupt 3 Interrupt Disable. See functional description for bit 0.
22	ICE_ADC	ADC Interrupt Disable. See functional description for bit 0.
23	ICE_BOD	BOD Interrupt Disable. See functional description for bit 0.
24	ICE_USB	USB Interrupt Disable. See functional description for bit 0.
25	ICE_CAN	CAN Interrupt Disable. See functional description for bit 0.
26	ICE_DMA	GPDMA Interrupt Disable. See functional description for bit 0.
27	ICE_I2S	I^2S Interrupt Disable. See functional description for bit 0.
28	ICE_ENET	Ethernet Interrupt Disable. See functional description for bit 0.
29	ICE_RIT	Repetitive Interrupt Timer Interrupt Disable. See functional description for bit 0.
30	ICE_MCPWM	Motor Control PWM Interrupt Disable. See functional description for bit 0.
31	ICE_QEI	Quadrature Encoder Interface Interrupt Disable. See functional description for bit 0.

6.5.4 Interrupt Clear-Enable Register 1 register (ICER1 - 0xE000 E184)

The ICER1 register allows disabling the second group of peripheral interrupts, or for reading the enabled state of those interrupts. Enabling interrupts is done through the ISER0 and ISER1 registers (Section 6.5.1 and Section 6.5.2).

Table 55. Interrupt Clear-Enable Register 1 register (ICER1 - 0xE000 E184)

Bit	Name	Function
0	ICE_PLL1	PLL1 (USB PLL) Interrupt Disable. Write: writing 0 has no effect, writing 1 disables the interrupt. Read: 0 indicates that the interrupt is disabled, 1 indicates that the interrupt is enabled.
1	ICE_USBACT	USB Activity Interrupt Disable. See functional description for bit 0.
2	ICE_CANACT	CAN Activity Interrupt Disable. See functional description for bit 0.
31:3	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.

6.5.5 Interrupt Set-Pending Register 0 register (ISPR0 - 0xE000 E200)

The ISPR0 register allows setting the pending state of the first 32 peripheral interrupts, or for reading the pending state of those interrupts. The remaining interrupts can have their pending state set via the ISPR1 register (Section 6.5.6). Clearing the pending state of interrupts is done through the ICPR0 and ICPR1 registers (Section 6.5.7 and Section 6.5.8).

Table 56. Interrupt Set-Pending Register 0 register (ISPR0 - 0xE000 E200)

Bit	Name	Function
0	ISP_WDT	Watchdog Timer Interrupt Pending set. Write: writing 0 has no effect, writing 1 changes the interrupt state to pending. Read: 0 indicates that the interrupt is not pending, 1 indicates that the interrupt is pending.
1	ISP_TIMER0	Timer 0 Interrupt Pending set. See functional description for bit 0.
2	ISP_TIMER1	Timer 1. Interrupt Pending set. See functional description for bit 0.
3	ISP_TIMER2	Timer 2 Interrupt Pending set. See functional description for bit 0.
4	ISP_TIMER3	Timer 3 Interrupt Pending set. See functional description for bit 0.
5	ISP_UART0	UART0 Interrupt Pending set. See functional description for bit 0.
6	ISP_UART1	UART1 Interrupt Pending set. See functional description for bit 0.
7	ISP_UART2	UART2 Interrupt Pending set. See functional description for bit 0.
8	ISP_UART3	UART3 Interrupt Pending set. See functional description for bit 0.
9	ISP_PWM	PWM1 Interrupt Pending set. See functional description for bit 0.
10	ISP_I2C0	I^2C0 Interrupt Pending set. See functional description for bit 0.
11	ISP_I2C1	I^2C1 Interrupt Pending set. See functional description for bit 0.
12	ISP_I2C2	I^2C2 Interrupt Pending set. See functional description for bit 0.
13	ISP_SPI	SPI Interrupt Pending set. See functional description for bit 0.
14	ISP_SSP0	SSP0 Interrupt Pending set. See functional description for bit 0.
15	ISP_SSP1	SSP1 Interrupt Pending set. See functional description for bit 0.
16	ISP_PLL0	PLL0 (Main PLL) Interrupt Pending set. See functional description for bit 0.
17	ISP_RTC	Real Time Clock (RTC) Interrupt Pending set. See functional description for bit 0.
18	ISP_EINT0	External Interrupt 0 Interrupt Pending set. See functional description for bit 0.
19	ISP_EINT1	External Interrupt 1 Interrupt Pending set. See functional description for bit 0.
20	ISP_EINT2	External Interrupt 2 Interrupt Pending set. See functional description for bit 0.
21	ISP_EINT3	External Interrupt 3 Interrupt Pending set. See functional description for bit 0.
22	ISP_ADC	ADC Interrupt Pending set. See functional description for bit 0.
23	ISP_BOD	BOD Interrupt Pending set. See functional description for bit 0.
24	ISP_USB	USB Interrupt Pending set. See functional description for bit 0.
25	ISP_CAN	CAN Interrupt Pending set. See functional description for bit 0.
26	ISP_DMA	GPDMA Interrupt Pending set. See functional description for bit 0.
27	ISP_I2S	I^2S Interrupt Pending set. See functional description for bit 0.
28	ISP_ENET	Ethernet Interrupt Pending set. See functional description for bit 0.
29	ISP_RIT	Repetitive Interrupt Timer Interrupt Pending set. See functional description for bit 0.
30	ISP_MCPWM	Motor Control PWM Interrupt Pending set. See functional description for bit 0.
31	ISP_QEI	Quadrature Encoder Interface Interrupt Pending set. See functional description for bit 0.

6.5.6 Interrupt Set-Pending Register 1 register (ISPR1 - 0xE000 E204)

The ISPR1 register allows setting the pending state of the second group of peripheral interrupts, or for reading the pending state of those interrupts. Clearing the pending state of interrupts is done through the ICPR0 and ICPR1 registers (Section 6.5.7 and Section 6.5.8).

Table 57. Interrupt Set-Pending Register 1 register (ISPR1 - 0xE000 E204)

Bit	Name	Function
0	ISP_PLL1	PLL1 (USB PLL) Interrupt Pending set. Write: writing 0 has no effect, writing 1 changes the interrupt state to pending. Read: 0 indicates that the interrupt is not pending, 1 indicates that the interrupt is pending.
1	ISP_USBACT	USB Activity Interrupt Pending set. See functional description for bit 0.
2	ISP_CANACT	CAN Activity Interrupt Pending set. See functional description for bit 0.
31:3	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.

6.5.7 Interrupt Clear-Pending Register 0 register (ICPR0 - 0xE000 E280)

The ICPR0 register allows clearing the pending state of the first 32 peripheral interrupts, or for reading the pending state of those interrupts. The remaining interrupts can have their pending state cleared via the ICPR1 register (Section 6.5.8). Setting the pending state of interrupts is done through the ISPR0 and ISPR1 registers (Section 6.5.5 and Section 6.5.6).

Table 58. Interrupt Clear-Pending Register 0 register (ICPR0 - 0xE000 E280)

Bit	Name	Function
0	ICP_WDT	Watchdog Timer Interrupt Pending clear. Write: writing 0 has no effect, writing 1 changes the interrupt state to not pending. Read: 0 indicates that the interrupt is not pending, 1 indicates that the interrupt is pending.
1	ICP_TIMER0	Timer 0 Interrupt Pending clear. See functional description for bit 0.
2	ICP_TIMER1	Timer 1. Interrupt Pending clear. See functional description for bit 0.
3	ICP_TIMER2	Timer 2 Interrupt Pending clear. See functional description for bit 0.
4	ICP_TIMER3	Timer 3 Interrupt Pending clear. See functional description for bit 0.
5	ICP_UART0	UART0 Interrupt Pending clear. See functional description for bit 0.
6	ICP_UART1	UART1 Interrupt Pending clear. See functional description for bit 0.
7	ICP_UART2	UART2 Interrupt Pending clear. See functional description for bit 0.
8	ICP_UART3	UART3 Interrupt Pending clear. See functional description for bit 0.
9	ICP_PWM	PWM1 Interrupt Pending clear. See functional description for bit 0.
10	ICP_I2C0	I^{2}C0 Interrupt Pending clear. See functional description for bit 0.
11	ICP_I2C1	I^{2}C1 Interrupt Pending clear. See functional description for bit 0.
12	ICP_I2C2	I^{2}C2 Interrupt Pending clear. See functional description for bit 0.
13	ICP_SPI	SPI Interrupt Pending clear. See functional description for bit 0.
14	ICP_SSP0	SSP0 Interrupt Pending clear. See functional description for bit 0.
15	ICP_SSP1	SSP1 Interrupt Pending clear. See functional description for bit 0.
16	ICP_PLL0	PLL0 (Main PLL) Interrupt Pending clear. See functional description for bit 0.
17	ICP_RTC	Real Time Clock (RTC) Interrupt Pending clear. See functional description for bit 0.
18	ICP_EINT0	External Interrupt 0 Interrupt Pending clear. See functional description for bit 0.
19	ICP_EINT1	External Interrupt 1 Interrupt Pending clear. See functional description for bit 0.
20	ICP_EINT2	External Interrupt 2 Interrupt Pending clear. See functional description for bit 0.
21	ICP_EINT3	External Interrupt 3 Interrupt Pending clear. See functional description for bit 0.
22	ICP_ADC	ADC Interrupt Pending clear. See functional description for bit 0.
23	ICP_BOD	BOD Interrupt Pending clear. See functional description for bit 0.
24	ICP_USB	USB Interrupt Pending clear. See functional description for bit 0.
25	ICP_CAN	CAN Interrupt Pending clear. See functional description for bit 0.
26	ICP_DMA	GPDMA Interrupt Pending clear. See functional description for bit 0.
27	ICP_I2S	I^{2}S Interrupt Pending clear. See functional description for bit 0.
28	ICP_ENET	Ethernet Interrupt Pending clear. See functional description for bit 0.
29	ICP_RIT	Repetitive Interrupt Timer Interrupt Pending clear. See functional description for bit 0.
30	ICP_MCPWM	Motor Control PWM Interrupt Pending clear. See functional description for bit 0.
31	ICP_QEI	Quadrature Encoder Interface Interrupt Pending clear. See functional description for bit 0.

6.5.8 Interrupt Clear-Pending Register 1 register (ICPR1 - 0xE000 E284)

The ICPR1 register allows clearing the pending state of the second group of peripheral interrupts, or for reading the pending state of those interrupts. Setting the pending state of interrupts is done through the ISPR0 and ISPR1 registers (Section 6.5.5 and Section 6.5.6).

Table 59. Interrupt Set-Pending Register 1 register (ISPR1 - 0xE000 E204)

Bit	Name	Function
0	ICP_PLL1	PLL1 (USB PLL) Interrupt Pending clear. Write: writing 0 has no effect, writing 1 changes the interrupt state to not pending. Read: 0 indicates that the interrupt is not pending, 1 indicates that the interrupt is pending.
1	ICP_USBACT	USB Activity Interrupt Pending clear. See functional description for bit 0.
2	ICP_CANACT	CAN Activity Interrupt Pending clear. See functional description for bit 0.
31:3	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.

6.5.9 Interrupt Active Bit Register 0 (IABR0 - 0xE000 E300)

The IABR0 register is a read-only register that allows reading the active state of the first 32 peripheral interrupts. This allows determining which peripherals are asserting an interrupt to the NVIC, and may also be pending if they are enabled. The remaining interrupts can have their active state read via the IABR1 register (Section 6.5.10).

Table 60. Interrupt Active Bit Register 0 (IABR0 - 0xE000 E300)

Bit	Name	Function
0	IAB_WDT	Watchdog Timer Interrupt Active. Read: 0 indicates that the interrupt is not active, 1 indicates that the interrupt is active.
1	IAB_TIMER0	Timer 0 Interrupt Active. See functional description for bit 0.
2	IAB_TIMER1	Timer 1. Interrupt Active. See functional description for bit 0.
3	IAB_TIMER2	Timer 2 Interrupt Active. See functional description for bit 0.
4	IAB_TIMER3	Timer 3 Interrupt Active. See functional description for bit 0.
5	IAB_UART0	UART0 Interrupt Active. See functional description for bit 0.
6	IAB_UART1	UART1 Interrupt Active. See functional description for bit 0.
7	IAB_UART2	UART2 Interrupt Active. See functional description for bit 0.
8	IAB_UART3	UART3 Interrupt Active. See functional description for bit 0.
9	IAB_PWM	PWM1 Interrupt Active. See functional description for bit 0.
10	IAB_I2C0	I^2C0 Interrupt Active. See functional description for bit 0.
11	IAB_I2C1	I^2C1 Interrupt Active. See functional description for bit 0.
12	IAB_I2C2	I^2C2 Interrupt Active. See functional description for bit 0.
13	IAB_SPI	SPI Interrupt Active. See functional description for bit 0.
14	IAB_SSP0	SSP0 Interrupt Active. See functional description for bit 0.
15	IAB_SSP1	SSP1 Interrupt Active. See functional description for bit 0.
16	IAB_PLL0	PLL0 (Main PLL) Interrupt Active. See functional description for bit 0.
17	IAB_RTC	Real Time Clock (RTC) Interrupt Active. See functional description for bit 0.
18	IAB_EINT0	External Interrupt 0 Interrupt Active. See functional description for bit 0.
19	IAB_EINT1	External Interrupt 1 Interrupt Active. See functional description for bit 0.
20	IAB_EINT2	External Interrupt 2 Interrupt Active. See functional description for bit 0.
21	IAB_EINT3	External Interrupt 3 Interrupt Active. See functional description for bit 0.
22	IAB_ADC	ADC Interrupt Active. See functional description for bit 0.
23	IAB_BOD	BOD Interrupt Active. See functional description for bit 0.
24	IAB_USB	USB Interrupt Active. See functional description for bit 0.
25	IAB_CAN	CAN Interrupt Active. See functional description for bit 0.
26	IAB_DMA	GPDMA Interrupt Active. See functional description for bit 0.
27	IAB_I2S	I^2S Interrupt Active. See functional description for bit 0.
28	IAB_ENET	Ethernet Interrupt Active. See functional description for bit 0.
29	IAB_RIT	Repetitive Interrupt Timer Interrupt Active. See functional description for bit 0.
30	IAB_MCPWM	Motor Control PWM Interrupt Active. See functional description for bit 0.
31	IAB_QEI	Quadrature Encoder Interface Interrupt Active. See functional description for bit 0.

6.5.10 Interrupt Active Bit Register 1 (IABR1 - 0xE000 E304)

The IABR1 register is a read-only register that allows reading the active state of the second group of peripheral interrupts. This allows determining which peripherals are asserting an interrupt to the NVIC, and may also be pending if they are enabled.

Table 61. Interrupt Active Bit Register 1 (IABR1 - 0xE000 E304)

Bit	Name	Function
0	IAB_PLL1	PLL1 (USB PLL) Interrupt Active. Read: 0 indicates that the interrupt is not active, 1 indicates that the interrupt is active.
1	IAB_USBACT	USB Activity Interrupt Active. See functional description for bit 0.
2	IAB_CANACT	CAN Activity Interrupt Active. See functional description for bit 0.
31:3	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.

6.5.11 Interrupt Priority Register 0 (IPR0 - 0xE000 E400)

The IPR0 register controls the priority of the first 4 peripheral interrupts. Each interrupt can have one of 32 priorities, where 0 is the highest priority.

Table 62. Interrupt Priority Register 0 (IPR0 - 0xE000 E400)

Bit	Name	Function
2:0	Unimplemented	These bits ignore writes, and read as 0.
7:3	IP_WDT	Watchdog Timer Interrupt Priority. 0 = highest priority. 31 (0x1F) = lowest priority.
10:8	Unimplemented	These bits ignore writes, and read as 0.
15:11	IP_TIMER0	Timer 0 Interrupt Priority. See functional description for bits 7-3.
18:16	Unimplemented	These bits ignore writes, and read as 0.
23:19	IP_TIMER1	Timer 1 Interrupt Priority. See functional description for bits 7-3.
26:24	Unimplemented	These bits ignore writes, and read as 0.
31:27	IP_TIMER2	Timer 2 Interrupt Priority. See functional description for bits 7-3.

6.5.12 Interrupt Priority Register 1 (IPR1 - 0xE000 E404)

The IPR1 register controls the priority of the second group of 4 peripheral interrupts. Each interrupt can have one of 32 priorities, where 0 is the highest priority.

Table 63. Interrupt Priority Register 1 (IPR1 - 0xE000 E404)

Bit	Name	Function
2:0	Unimplemented	These bits ignore writes, and read as 0.
7:3	IP_TIMER3	Timer 3 Interrupt Priority. 0 = highest priority. 31 (0x1F) = lowest priority.
10:8	Unimplemented	These bits ignore writes, and read as 0.
15:11	IP_UART0	UART0 Interrupt Priority. See functional description for bits 7-3.
18:16	Unimplemented	These bits ignore writes, and read as 0.
23:19	IP_UART1	UART1 Interrupt Priority. See functional description for bits 7-3.
26:24	Unimplemented	These bits ignore writes, and read as 0.
31:27	IP_UART2	UART2 Interrupt Priority. See functional description for bits 7-3.

6.5.13 Interrupt Priority Register 2 (IPR2 - 0xE000 E408)

The IPR2 register controls the priority of the third group of 4 peripheral interrupts. Each interrupt can have one of 32 priorities, where 0 is the highest priority.

Table 64. Interrupt Priority Register 2 (IPR2 - 0xE000 E408)

Bit	Name	Function
2:0	Unimplemented	These bits ignore writes, and read as 0.
7:3	IP_UART3	UART3 Interrupt Priority. 0 = highest priority. 31 (0x1F) = lowest priority.
10:8	Unimplemented	These bits ignore writes, and read as 0.
15:11	IP_PWM	PWM Interrupt Priority. See functional description for bits 7-3.
18:16	Unimplemented	These bits ignore writes, and read as 0.
23:19	IP_I2C0	I^2C0 Interrupt Priority. See functional description for bits 7-3.
26:24	Unimplemented	These bits ignore writes, and read as 0.
31:27	IP_I2C1	I^2C1 Interrupt Priority. See functional description for bits 7-3.

6.5.14 Interrupt Priority Register 3 (IPR3 - 0xE000 E40C)

The IPR3 register controls the priority of the fourth group of 4 peripheral interrupts. Each interrupt can have one of 32 priorities, where 0 is the highest priority.

Table 65. Interrupt Priority Register 3 (IPR3 - 0xE000 E40C)

Bit	Name	Function
2:0	Unimplemented	These bits ignore writes, and read as 0.
7:3	IP_I2C2	I^2C2 Interrupt Priority. 0 = highest priority. 31 (0x1F) = lowest priority.
10:8	Unimplemented	These bits ignore writes, and read as 0.
15:11	IP_SPI	SPI Interrupt Priority. See functional description for bits 7-3.
18:16	Unimplemented	These bits ignore writes, and read as 0.
23:19	IP_SSP0	SSP0 Interrupt Priority. See functional description for bits 7-3.
26:24	Unimplemented	These bits ignore writes, and read as 0.
31:27	IP_SSP1	SSP1 Interrupt Priority. See functional description for bits 7-3.

6.5.15 Interrupt Priority Register 4 (IPR4 - 0xE000 E410)

The IPR4 register controls the priority of the fifth group of 4 peripheral interrupts. Each interrupt can have one of 32 priorities, where 0 is the highest priority.

Table 66. Interrupt Priority Register 4 (IPR4 - 0xE000 E410)

Bit	Name	Function
2:0	Unimplemented	These bits ignore writes, and read as 0.
7:3	IP_PLL0	PLL0 (Main PLL) Interrupt Priority. 0 = highest priority. 31 (0x1F) = lowest priority.
10:8	Unimplemented	These bits ignore writes, and read as 0.
15:11	IP_RTC	Real Time Clock (RTC) Interrupt Priority. See functional description for bits 7-3.
18:16	Unimplemented	These bits ignore writes, and read as 0.
23:19	IP_EINT0	External Interrupt 0 Interrupt Priority. See functional description for bits 7-3.
26:24	Unimplemented	These bits ignore writes, and read as 0.
31:27	IP_EINT1	External Interrupt 1 Interrupt Priority. See functional description for bits 7-3.

6.5.16 Interrupt Priority Register 5 (IPR5 - 0xE000 E414)

The IPR5 register controls the priority of the sixth group of 4 peripheral interrupts. Each interrupt can have one of 32 priorities, where 0 is the highest priority.

Table 67. Interrupt Priority Register 5 (IPR5 - 0xE000 E414)

Bit	Name	Function
2:0	Unimplemented	These bits ignore writes, and read as 0.
7:3	IP_EINT2	External Interrupt 2 Interrupt Priority. 0 = highest priority. 31 (0x1F) = lowest priority.
10:8	Unimplemented	These bits ignore writes, and read as 0.
15:11	IP_EINT3	External Interrupt 3 Interrupt Priority. See functional description for bits 7-3.
18:16	Unimplemented	These bits ignore writes, and read as 0.
23:19	IP_ADC	ADC Interrupt Priority. See functional description for bits 7-3.
26:24	Unimplemented	These bits ignore writes, and read as 0.
31:27	IP_BOD	BOD Interrupt Priority. See functional description for bits 7-3.

6.5.17 Interrupt Priority Register 6 (IPR6 - 0xE000 E418)

The IPR6 register controls the priority of the seventh group of 4 peripheral interrupts. Each interrupt can have one of 32 priorities, where 0 is the highest priority.

Table 68. Interrupt Priority Register 6 (IPR6 - 0xE000 E418)

Bit	Name	Function
2:0	Unimplemented	These bits ignore writes, and read as 0.
7:3	IP_USB	USB Interrupt Priority. 0 = highest priority. 31 (0x1F) = lowest priority.
10:8	Unimplemented	These bits ignore writes, and read as 0.
15:11	IP_CAN	CAN Interrupt Priority. See functional description for bits 7-3.
18:16	Unimplemented	These bits ignore writes, and read as 0.
23:19	IP_DMA	GPDMA Interrupt Priority. See functional description for bits 7-3.
26:24	Unimplemented	These bits ignore writes, and read as 0.
31:27	IP_I2S	I^2S Interrupt Priority. See functional description for bits 7-3.

6.5.18 Interrupt Priority Register 7 (IPR7 - 0xE000 E41C)

The IPR7 register controls the priority of the eighth group of 4 peripheral interrupts. Each interrupt can have one of 32 priorities, where 0 is the highest priority.

Table 69. Interrupt Priority Register 7 (IPR7 - 0xE000 E41C)

Bit	Name	Function
2:0	Unimplemented	These bits ignore writes, and read as 0.
7:3	IP_ENET	Ethernet Interrupt Priority. 0 = highest priority. 31 (0x1F) = lowest priority.
10:8	Unimplemented	These bits ignore writes, and read as 0.
15:11	IP_RIT	Repetitive Interrupt Timer Interrupt Priority. See functional description for bits 7-3.
18:16	Unimplemented	These bits ignore writes, and read as 0.
23:19	IP_MCPWM	Motor Control PWM Interrupt Priority. See functional description for bits 7-3.
26:24	Unimplemented	These bits ignore writes, and read as 0.
31:27	IP_QEI	Quadrature Encoder Interface Interrupt Priority. See functional description for bits 7-3.

6.5.19 Interrupt Priority Register 8 (IPR8 - 0xE000 E420)

The IPR8 register controls the priority of the ninth and last group of 4 peripheral interrupts. Each interrupt can have one of 32 priorities, where 0 is the highest priority.

Table 70. Interrupt Priority Register 8 (IPR8 - 0xE000 E420)

Bit	Name	Function
2:0	Unimplemented	These bits ignore writes, and read as 0.
7:3	IP_PLL1	PLL1 (USB PLL) Interrupt Priority. 0 = highest priority. 31 (0x1F) = lowest priority.
10:8	Unimplemented	These bits ignore writes, and read as 0.
15:11	IP_USBACT	USB Activity Interrupt Priority. See functional description for bits 7-3.
18:16	Unimplemented	These bits ignore writes, and read as 0.
23:19	IP_CANACT	CAN Activity Interrupt Priority. See functional description for bits 7-3.
31:24	Unimplemented	These bits ignore writes, and read as 0.

6.5.20 Software Trigger Interrupt Register (STIR - 0xE000 EF00)

The STIR register provides an alternate way for software to generate an interrupt, in addition to using the ISPR registers. This mechanism can only be used to generate peripheral interrupts, not system exceptions.

By default, only privileged software can write to the STIR register. Unprivileged software can be given this ability if privileged software sets the USERSETMPEND bit in the CCR register (see Section 34.4.3.8).

Table 71. Software Trigger Interrupt Register (STIR - 0xE000 EF00)

Bit	Name	Function
8:0	INTID	Writing a value to this field generates an interrupt for the specified the interrupt number (see Table 50). The range allowed for the LPC176x/5x is 0 to 111.
31:9	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.